

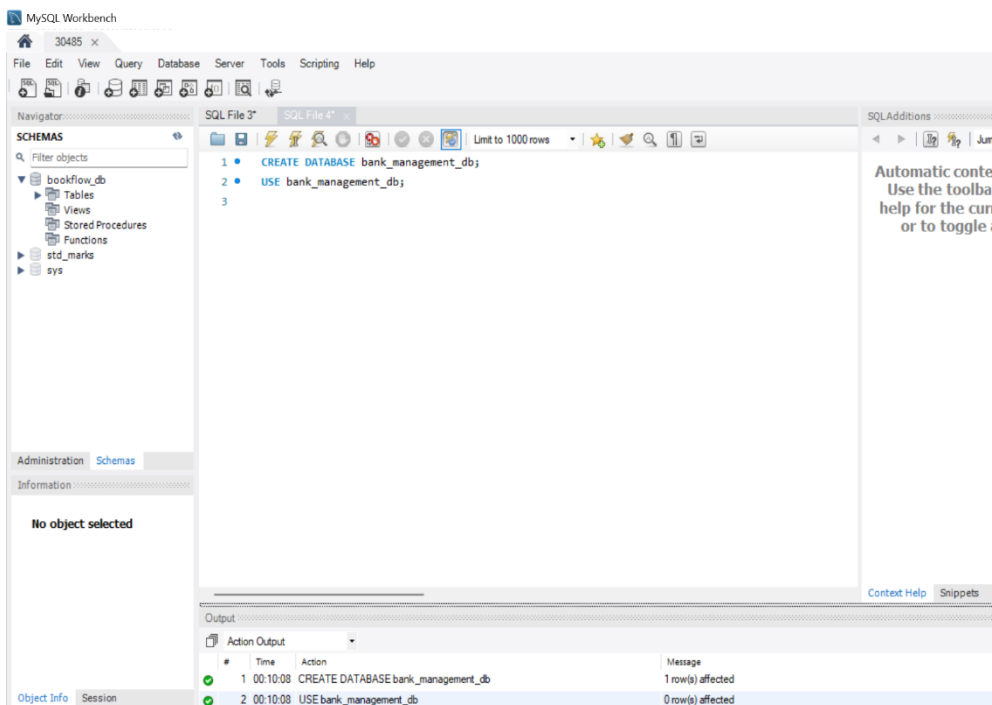
Commands

M. Geethika (2520030485)

Question :

Perform the following DDL, DML, DCL, and TCL operations on a Bank Management System database: create and alter a bank transaction table, add and modify columns, rename, truncate and drop tables; insert, update, delete, and retrieve transaction records using SELECT, WHERE, and ORDER BY; grant and revoke user privileges; and perform transaction control operations using COMMIT, SAVEPOINT, and ROLLBACK.

Step 1 — Create Database



Step 2 — DDL Commands

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - std_marks
 - sys

Administration Schemas

Information

No object selected

SQL File 3*

SQL File 4*

Limit to 1000 rows

```

1 • CREATE DATABASE bank_management_db;
2 • USE bank_management_db;
3 -- Step 2: Create Bank Transactions Table
4 • CREATE TABLE bank_transactions(
5     txn_id INT PRIMARY KEY,
6     customer_name VARCHAR(50),
7     branch_name VARCHAR(50),
8     transaction_type VARCHAR(20),
9     amount DECIMAL(10,2),
10    transaction_date DATE
11 );

```

SQL Additions

Autosave help

Output

Action Output

#	Time	Action	Message
1	00:10:08	CREATE DATABASE bank_management_db	1 row(s) affected
2	00:10:08	USE bank_management_db	0 row(s) affected
3	00:10:08	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_na...	0 row(s) affected

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bookflow_db
 - Tables
 - Views
 - Stored Procedures
 - Functions
 - std_marks
 - sys

Administration Schemas

Information

No object selected

SQL File 3*

SQL File 4*

Limit to 1000 rows

```

4 • CREATE TABLE bank_transactions(
5     txn_id INT PRIMARY KEY,
6     customer_name VARCHAR(50),
7     branch_name VARCHAR(50),
8     transaction_type VARCHAR(20),
9     amount DECIMAL(10,2),
10    transaction_date DATE
11 );
12 -- Step 3: Add Account Number Column
13 • ALTER TABLE bank_transactions
14     ADD account_no VARCHAR(20);
15 -- Step 4: Modify Customer Name Column
16 • ALTER TABLE bank_transactions
17     MODIFY customer_name VARCHAR(100);
18 -- Step 5: Rename Transaction Table
19 • RENAME TABLE bank_transactions
20     TO customer_transactions;

```

SQL Additions

Automatic content
Use the toolbar
help for the current
or to toggle

Context Help Snippets

Output

Action Output

#	Time	Action	Message
1	00:10:08	CREATE DATABASE bank_management_db	1 row(s) affected
2	00:10:08	USE bank_management_db	0 row(s) affected
3	00:10:08	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_na...	0 row(s) affected
4	00:10:08	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	00:10:08	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
6	00:10:08	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

Object Info Session

File Edit View Query Database Server Tools Scripting Help

SQL File 3* SQL File 4* x

Limit to 1000 rows

SQL File 3* SQL File 4* x

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	00:10:08	CREATE DATABASE bank_management_db	1 row(s) affected	0.015 sec
2	00:10:08	USE bank_management_db	0 row(s) affected	0.000 sec
3	00:10:08	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_na...	0 row(s) affected	0.078 sec
4	00:10:08	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.203 sec
5	00:10:08	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.094 sec
6	00:10:08	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.031 sec
7	00:10:08	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.063 sec
8	00:10:08	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected	0.031 sec
9	00:10:08	DROP TABLE bank_backup	0 row(s) affected	0.031 sec

Step 3: DML Commands

```
24  -- Step 7: Create Backup Table
25  • CREATE TABLE bank_backup(
26      txn_id INT,
27      customer_name VARCHAR(100)
28  );
29  -- Step 8: Drop Backup Table
30  • DROP TABLE bank_backup;
```

Output

#	Time	Action	Message
3	12:56:01	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
4	12:57:01	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
8	13:06:40	DROP TABLE bank_backup	0 row(s) affected

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

bookflow_db

Tables

Views

Stored Procedures

Functions

std_marks

sys

Administration Schemas

Information

No object selected

Object Info Session

SQL File 3*

SQL File 4*

Limit to 1000 rows

SQLAdditions

Automatic context help
Use the toolbar to ma
help for the current ca
or to toggle autom

Context Help Snippets

```

30 -- Step 9: Insert Sample Data
31 • INSERT INTO customer_transactions
32 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
33 VALUES
34 (101, 'Geethika', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
35 (102, 'Shailu', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
36 (103, 'Siri', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
37 (104, 'Jhansi', 'Vizag', 'Deposit', 8000, '2024-01-10'),
38 (105, 'Bhavitha', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
39 (106, 'Nakshatra', 'Vizag', 'Deposit', 15000, '2024-01-12'),
40 (107, 'Pavani', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
41 (108, 'Aishu', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
42 (109, 'Mahija', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
43 (110, 'Praksh', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
44 -- Step 10: Insert New Transaction
45 • INSERT INTO customer_transactions
46 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
47 VALUES

```

Output

Action Output

#	Time	Action	Message	D
7	00:10:08	TRUNCATE TABLE customer_transactions	0 row(s) affected	0.0
8	00:10:08	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected	0.0
9	00:10:08	DROP TABLE bank_backup	0 row(s) affected	0.0
10	00:10:08	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, tran...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.0
11	00:10:08	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, tran...	1 row(s) affected	0.0

```

45 -- Step 10: Insert New Transaction
46 • INSERT INTO customer_transactions
47 (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
48 VALUES
49 (111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');

```

Output

Action Output

#	Time	Action	Message
5	12:57:58	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected
6	13:04:15	TRUNCATE TABLE customer_transactions	0 row(s) affected
7	13:05:13	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
8	13:06:40	DROP TABLE bank_backup	0 row(s) affected
9	13:09:40	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
10	13:11:46	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transa...	1 row(s) affected

MySQL Workbench

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bank_management_db

Tables could not be fetched

Views could not be fetched

Stored Procedures could not be fetched

Functions could not be fetched

bookflow_db

std_marks

sys

Query 1

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

148

149 -- Final Display

150 • SELECT * FROM customer_transactions;

151

152 -- Check Available Tables

153 • SHOW TABLES;

154

Result Grid

Filter Rows:

bn_id customer_name branch_name transaction_type amount transaction_date account_no

101	Geethika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Siri	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Jhansi	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Shailu	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Bhavitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Nakshh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Yamini	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Mahija	Hyderabad	Deposit	9000.00	2024-01-14	NULL

customer_transactions 6 x customer_transactions 7 customer_transactions 8 customer_transactions 9 Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
79	16:40:43	UPDATE customer_transactions SET amount = 9000 WHERE bn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
80	16:40:43	ROLLBACK TO SAVEPOINT SP1	0 row(s) affected	0.000 sec
81	16:40:43	COMMIT	0 row(s) affected	0.000 sec
82	16:40:43	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
83	16:40:43	SHOW TABLES	1 row(s) returned	0.000 sec / 0.000 sec

Schema: bank_management_db

Object Info Session

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bank_management_db

Tables could not be fetched

Views could not be fetched

Stored Procedures could not be fetched

Functions could not be fetched

bookflow_db

std_marks

sys

Query 1 x

Limit to 1000 rows

148

149 -- Final Display

150 • SELECT * FROM customer_transactions;

151

152 -- Check Available Tables

153 • SHOW TABLES;

154

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Geethika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Jhansi	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Shailu	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Nakshh	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Mahija	Hyderabad	Vizag	9000.00	2024-01-14	NULL
110	Diksha	Vijayawada	Deposit	11000.00	2024-01-16	NULL

customer_transactions 6 customer_transactions 7 x customer_transactions 8 customer_transactions 9 Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message
79	16:40:43	UPDATE customer_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
80	16:40:43	ROLLBACK TO SAVEPOINT SP1	0 row(s) affected
81	16:40:43	COMMIT	0 row(s) affected
82	16:40:43	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
83	16:40:43	SHOW TABLES	1 row(s) returned

Object Info Session

Automatic context help
Use the toolbar to help for the current object or to toggle automatic context help

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bank_management_db

Tables could not be fetched

Views could not be fetched

Stored Procedures could not be fetched

Functions could not be fetched

bookflow_db

std_marks

sys

Query 1 x

Limit to 1000 rows

148

149 -- Final Display

150 • SELECT * FROM customer_transactions;

151

152 -- Check Available Tables

153 • SHOW TABLES;

154

Result Grid

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Geethika	Hyderabad	Deposit	7000.00	2024-01-05	NULL
102	Siri	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Jhansi	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Shailu	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Bhavitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Nakshh	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Yamini	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL
108	Mahija	Hyderabad	Deposit	9000.00	2024-01-14	NULL

customer_transactions 6 customer_transactions 7 customer_transactions 8 customer_transactions 9 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message
79	16:40:43	UPDATE customer_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
80	16:40:43	ROLLBACK TO SAVEPOINT SP1	0 row(s) affected
81	16:40:43	COMMIT	0 row(s) affected
82	16:40:43	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
83	16:40:43	SHOW TABLES	1 row(s) returned

Object Info Session

Automatic context help
Use the toolbar to help for the current object or to toggle automatic context help

Step 4: DCL Commands

The screenshot displays the SQL Developer interface with the following components:

- Navigator:** Shows the database schema structure, including `bank_management_db` (Tables, Views, Stored Procedures, Functions), `bookflow_db`, `std_marks`, and `sys`.
- Commands:** Contains the following SQL commands:

```
79 WHERE transaction_type = 'Deposit';
80
81 -- Step 15: Sort Transactions by Amount
82 • SELECT * FROM customer_transactions
83   ORDER BY amount DESC;
84
85 -- Create Auditor1 User
86 • CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'
87   IDENTIFIED BY '1234';
88
89 -- Create BankManager User
```
- Output:** Shows the execution results of the commands, including the number of rows affected and returned.
- Schema:** Displays the current schema, `bank_management_db`.

#	Time	Action	Message
✓ 18	16:46:04	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 19	16:46:04	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected
✓ 20	16:46:04	SELECT * FROM customer_transactions LIMIT 0, 1000	10 row(s) returned
✓ 21	16:46:04	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned
✓ 22	16:46:04	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned

30485 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bank_management_db

Tables

Views

Stored Procedures

Functions

bookflow_db

std_marks

sys

Administration Schemas

Information

Schema: bank_management_db

Commands

Limit to 1000 rows

```
93 -- Step 16: Grant SELECT Permission
94 GRANT SELECT ON bank_management_db.customer_transactions
95 TO 'Auditor1'@'localhost';
96
97 -- Step 17: Grant All Privileges
98 GRANT ALL PRIVILEGES ON bank_management_db.customer_transactions
99 TO 'BranchManager'@'localhost';
100
101 -- Step 18: Revoke SELECT Permission
102 REVOKE SELECT ON bank_management_db.customer_transactions
103 FROM 'Auditor1'@'localhost';
104
105 -- Step 19: Revoke All Privileges
106 REVOKE ALL PRIVILEGES ON bank_management_db.customer_transactions
107 FROM 'BranchManager'@'localhost';
108
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
21	16:46:04	SELECT * FROM customer_transactions WHERE transaction_type = 'Deposit' LIMIT...	6 row(s) returned	0.000 sec / 0.0
22	16:46:04	SELECT * FROM customer_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.0
23	16:46:04	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected, 1 warning(s): 3163 Authorization ID 'Auditor1'@'localhost' already ...	0.016 sec
24	16:46:04	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY '12...	0 row(s) affected, 1 warning(s): 3163 Authorization ID 'BranchManager'@'localhost' ...	0.016 sec
25	16:46:04	GRANT SELECT ON bank_management_db.customer_transactions TO 'Auditor1'...	0 row(s) affected	0.000 sec
26	16:46:04	GRANT ALL PRIVILEGES ON bank_management_db.customer_transactions TO '...	0 row(s) affected	0.015 sec
27	16:46:04	REVOKE SELECT ON bank_management_db.customer_transactions FROM 'Auditor1'...	0 row(s) affected	0.000 sec

Step 5: TCL Commands

The screenshot displays a database management interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar shows a 'SCHEMAS' tree with 'bank_management_db' selected. The main 'Commands' window contains the following SQL script:

```
109 -- Step 20: Commit Transaction
110 START TRANSACTION;
111
112 UPDATE customer_transactions
113 SET amount = 2000
114 WHERE txn_id = 102;
115
116 COMMIT;
117
118 -- Step 21: Create Savepoint
119 START TRANSACTION;
120
121 SAVEPOINT Before_Update;
122
123 -- Step 22: Update and Rollback to Savepoint
124 UPDATE customer_transactions
125 SET amount = 9999
126 WHERE txn_id = 102;
```

The right sidebar shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help." Below this is a 'Context Help' tab.

The 'Output' window at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
34	16:46:04	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
35	16:46:04	ROLLBACK TO SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
36	16:46:04	COMMIT	0 row(s) affected	0.000 sec
37	16:46:04	START TRANSACTION	0 row(s) affected	0.000 sec
38	16:46:04	UPDATE customer_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
39	16:46:04	SAVEPOINT SP1	0 row(s) affected	0.000 sec
40	16:46:04	UPDATE customer_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

